

SPACE DEBRIS GENERATION AND ORBITAL PROPAGATION PREDICTION USING AI/ML

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Introduction

Company Overview

Partner: ERETS (European Research Establishment for the Sustainable Space)

Our client for this project is ERETS Space, a private organization dedicated to space research with their HQ in London, UK. A multidisciplinary team of engineers specializing in their respective fields come together as ERETS to integrate AI, optics, autonomous robotics, and advanced propulsion systems to tackle one of the most pressing challenges in space—space debris. The company is developing technologies for eliminating space debris from Low Earth Orbit (LEO) and aims to de-orbit objects measuring 1cm – 10cm, so it can burn through our atmosphere. The organization's mission takes humanity one step closer towards sustainable space exploration, which happens to be a critical concern in the New Space age.

Business Objectives

NASA defines space debris as "any man-made object in orbit about the Earth which no longer serves a useful function" (2020, cited in Miller, 2021). The origin of this space debris can come from various sources; it can result from the sudden breakup of a satellite in orbit or an accidental collision. However, the biggest source of space debris comes from anti-satellite tests (Miller, 2021). The United States Space Force stated that due to the increasing amount of space debris, debris-related collisions will become more widespread, which will further affect orbital flight safety (2020, cited in Miller, 2021). In fact, the fact that space debris represents a significant risk to satellite operation was also mentioned in research by Lewis, Swinerd, and Newland (2011), which means that over the last 10 years, there have not been many changes made to address the problem. According to the European Space Agency (ESA), there are more than 29,000 pieces of space debris larger than 10 cm, 670,000 larger than 1 cm, and more than 170 million larger than 1 mm. These facts are robust evidence that this is a growing and urgent problem.

The main objective of the business is in the market of space debris management, ranging from space debris mitigation to tracking and predictions for events using DS/AI techniques. These operations will be carried out by two of their engineered satellites: ERETS-C satellite and ERETS-D. ERETS-C is a satellite constellation specified for debris tracking and detection in Low Earth Orbit (LEO) and Medium Earth Orbit (MEO)

AI and DS Application

ERETS Space believes that Data Science (DS) and Artificial Intelligence (AI) can help solve their problems for the following reason: Traditional prediction methods, reliant on historical data and physical modelling, often fall short in accurately forecasting debris generation, propagation, and its potential impacts. AI and ML techniques can predict space debris propagation, decay and re-entry data more effectively based on real-time debris generation events recorded in past decades (like anti-satellite missile tests or debris and satellite collisions).

Project Aims and the Final Deliverable

The initial aim of this project was to develop an AI/ML-based system that could predict and analyze space debris generation and orbital propagation patterns. This was to enhance the current methods of tracking and predicting space debris behavior, which primarily rely on historical data and physical modeling. We structured our approach in two main phases: first, creating a comprehensive learning database by collecting and processing historical space debris data, orbital parameters, and decay rates; and second, developing an intelligent predictive pipeline capable of forecasting debris trajectories, decay rates, and potential collision events.

As the project progressed, we encountered significant challenges in developing mathematical models due to the complexity of the domain knowledge required for space debris propagation. Additionally, our investigation revealed that publicly available data, particularly regarding object mass and radar cross section (RCS), was insufficient to develop a robust predictive model. Based on these constraints, we determined that pursuing this path would not be feasible within the project's scope and timeline.

Therefore, this project evolved into a comprehensive case study that provides valuable strategic insights for our partner. Through systematic investigation and testing of various model architectures - from basic regression to Multi-Layer Perceptron (MLP) models - we've documented both the current capabilities and limitations in AI-based debris prediction. Our findings clearly map out what information is currently available and, crucially, identify specific data gaps that must be addressed to build effective AI prediction systems. This knowledge enables our partner to make informed decisions about resource allocation and establish realistic timelines for future space debris monitoring projects. While the project's outcomes differ from our initial technical ambitions, the insights gained provide a solid foundation for understanding the complex challenges involved in space debris monitoring and guide the development of more focused, achievable approaches to this critical problem.

Methodology

Data Collection

The data collection and processing methodology for this project utilizes direct interaction with the Space-Track.org API through a custom-developed Python script. After initial attempts at web browser scraping ran into several limitations - proving to be time-consuming, producing inaccurate data, and being blocked by the site after a few hundred runs - we switched to using the API. This approach ensures efficient data retrieval while avoiding common limitations such as IP blocking and processing inefficiencies.

Our data collection process targets two essential datasets from Space-Track.org: the Satellite Catalog (SATCAT) and the General Perturbations (GP) dataset. The SATCAT provides fundamental information about space objects, while the GP dataset contains detailed orbital parameters. The implemented Python script handles secure authentication using environment variables and employs optimized HTTP requests to ensure efficient server interaction.

The data integration process combines the SATCAT and GP datasets using the NORAD Catalog ID (NORAD_CAT_ID) as the linking key. This integration creates comprehensive records for each space object, ensuring all orbital and object characteristics are properly consolidated in a single dataset. To

maximize efficiency, the implementation includes an incremental update mechanism. The script first checks for existing data files and identifies the highest existing NORAD_CAT_ID. This allows for selective retrieval of only new data entries, significantly reducing processing overhead while maintaining dataset completeness.

This methodology provides a robust framework for data collection and processing, establishing a foundation for analyzing orbital debris patterns and characteristics. The modular design allows for future enhancements, such as incorporating additional data sources or implementing new processing algorithms, while maintaining efficient and reliable data collection operations.

The automated nature of the system ensures consistent data collection while minimizing manual intervention, creating a reliable dataset for orbital debris analysis and tracking. This approach supports efficient data management and enables comprehensive analysis of space object characteristics and orbital patterns.

Data Loading and Preprocessing

The dataset was initially loaded from a JSON file that contained comprehensive satellite data, including essential metadata such as 'SATNAME' and 'COUNTRY', along with nested structures like 'omm_data'. These nested structures required special handling to extract the necessary information for analysis. To tackle this, we employed the `'pd.json_normalize()'` function, which allowed us to flatten these nested fields effectively. By doing so, we extracted subfields such as 'mean_motion' and 'eccentricity' from 'omm_data', organizing them into separate columns. This normalization process was crucial for structuring the data in a way that facilitated further analysis and processing.

In the data cleaning phase, we focused on managing missing values, particularly in critical columns like 'DECAY' and 'omm_data'. These columns are vital for accurate analysis, so it was essential to address any gaps. We handled missing data by filtering out records with null values, ensuring that only complete and reliable records were processed in subsequent steps. This approach helped maintain the integrity and quality of the dataset.

After extracting data from 'omm_data', we encountered several duplicated columns within the dataset. Some of these columns had different names but contained identical values, leading to redundancy. To resolve this issue, we conducted a thorough examination of the dataset. This involved manually inspecting the data and visualizing the correlation matrix to identify columns with identical data but different names. Based on this manual and visual analysis, we removed redundant columns, ensuring that each column represented unique and non-redundant information.

As a final step in data pre-processing, we calculated the satellite's lifetime by determining the difference between the 'DECAY' and 'LAUNCH' dates. This calculation resulted in a new column named 'LIFETIME_DAYS', which represents the number of days between a satellite's launch and its decay. This metric is crucial for prediction tasks related to satellite lifetime. To ensure accuracy in these calculations, both 'DECAY' and 'LAUNCH' dates were converted to datetime format. This conversion allowed us to compute the difference accurately, providing numerical values for 'LIFETIME_DAYS' that are suitable for use in machine learning models.

By following these steps, we ensured the dataset was structured, clean, and enriched with relevant features like '*LIFETIME_DAYS*' for better prediction and analysis outcomes.

Modelling

We trained and built all of our models entirely on Google Colab with the free T4 GPU. Given the high dimensionality of our scraped dataset and the complex interrelationships among variables, we implemented Principal Component Analysis (PCA) for feature engineering and dimensionality reduction (Figure 1 in Appendix). Based on the PCA results, we selected seven key features: apogee, orbital period, perigee, radar cross-section (RCS) size, object type, inclination, and ephemeris type. These features formed the basis for all models except the neural networks. Furthermore, we approached the decay date prediction through both regression and classification frameworks.

Regression Approach

In the regression approach, we evaluated multiple models, including linear and nonlinear methods, with their comparative performance metrics presented in Table 1. Random Forest Regression emerged as the best-performing model, achieving an RMSE of 2,435.94 days (approximately 6.67 years).

To address the substantial prediction error, we explored neural network architectures, beginning with a Multi-Layer Perceptron (MLP) Regressor. The base architecture comprised three hidden layers (200, 100, and 50 neurons) designed to progressively refine complex space environment patterns into decay predictions. The larger initial layer was implemented to capture the multitude of variables affecting satellite decay, while subsequent layers distilled these patterns into more focused decay-relevant features. To mitigate overfitting and optimize generalization, we incorporated:

- Early stopping with a 15% validation split
- Adaptive learning rate initialization at 0.001
- ReLU activation functions for non-linear relationship modeling
- Adam optimizer for training efficiency
- Maximum iteration limit of 10,000 with warm starts
- RobustScaler for feature normalization to handle outliers

This base model achieved an RMSE of 2,820.06 days (about 7 and a half years).

To enhance prediction accuracy, we developed a neural network using TensorFlow 2. The optimized architecture consisted of:

1. First dense layer: 96 neurons with ReLU activation, followed by 10% dropout
2. Second dense layer: 48 neurons with ReLU activation, no dropout
3. Output layer: Single neuron for final prediction

The model utilized the Adam optimizer with a learning rate of 0.0017, optimizing for Mean Squared Error (MSE) while monitoring both MSE and Mean Absolute Error (MAE). This architecture employs selective regularization through first-layer dropout, facilitating robust feature extraction followed by complete

pattern preservation in the second layer. The model achieved an RMSE of 2,493.90 days (about 7 years), with the loss epoch progression illustrated in Figure 2 in Appendix. Through subsequent evaluation and consultation with domain experts in the partner firm, we determined that the available dataset's limitations (RCS size and mass) constrained further significant improvements in regression performance. This led us to explore classification approaches as an alternative methodology.

Classification Approach

Following our regression analysis, we reformulated the decay date prediction as a classification problem, categorizing space debris into three lifetime classes: Short-lived, Medium-lived, and Long-lived objects. We evaluated four classical machine learning algorithms: Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Logistic Regression, using the same feature set identified through our earlier PCA analysis.

Results, Evaluation & Analysis

The project yielded several significant outcomes across data collection, preprocessing, and modeling phases. The results of our modeling efforts encompassing both regression and classification approaches are outlined below:

Regression Models

- Random Forest Regression emerged as the best performer among traditional models, achieving an RMSE of 2,335.84 days (approximately 6.4 years) and an R-squared value of 0.7134.
- A custom neural network implemented in TensorFlow achieved an RMSE of 2,493.90 days (about 6.8 years), slightly underperforming the Random Forest model.

Model	RMSE	R-squared
Random Forest	2,335.84	0.7134
Gradient Boosting	2,413.87	0.6940
K-Nearest Neighbors	2,536.94	0.6620
Elastic Net	3,776.66	0.2509
Ridge	3,776.95	0.2508
Lasso	3,776.95	0.2508
Polynomial	3,955.77	0.1781

Table 1: Regression Model Performance Comparison

Classification Models

The Random Forest classifier emerged as the most effective model, achieving an accuracy of 79.34%. This model demonstrated robust performance across classes, with particularly strong recall rates for Long-lived

(87%) and Short-lived (88%) objects. The K-Nearest Neighbors algorithm performed comparably well, achieving 78.47% accuracy with more balanced predictions across all classes, including notably better performance (58% recall) on the challenging Medium-lived category.

The Support Vector Machine and Logistic Regression models achieved accuracies of 68.86% and 64.27% respectively. While both models showed strength in identifying Long-lived (83-84% recall) and Short-lived (81-86% recall) objects, they struggled significantly with Medium-lived classifications. The Logistic Regression model, in particular, showed extreme bias against the Medium-lived class, with only 1% recall, effectively reducing the problem to binary classification.

A consistent pattern emerged across all models: the Medium-lived class proved most challenging to predict accurately. This difficulty likely stems from several factors:

1. Class imbalance in the training data
2. Potentially ambiguous boundaries between lifetime categories
3. Greater complexity in the decay patterns of objects in this intermediate lifetime range

Table 2 presents the accuracy comparison across all four models, illustrating the clear performance advantage of ensemble (Random Forest) and instance-based (KNN) methods over other approaches. These results suggest that the complex, nonlinear relationships in space debris decay patterns are better captured by models that can accommodate multiple decision boundaries and local pattern recognition.

Model	Accuracy in %
Random Forest	79.34
SVM	68.86
KNN	78.47
Logistic Regression	64.27

Table 2: Classification Model Performance Comparison

Based on these findings, we recommend the Random Forest classifier as the primary model for operational use, with KNN as a viable alternative when computational resources permit maintaining the full training dataset in memory. To further improve performance, particularly for Medium-lived objects, future work should consider:

- Implementing balanced sampling techniques or class weights
- Exploring ensemble methods combining multiple classifiers
- Collecting additional data points for Medium-lived objects to address class imbalance

The classification approach, while not eliminating prediction uncertainty, provides more reliable categorical predictions compared to the precise temporal estimates attempted in our regression models. This suggests that for practical space debris management, categorical lifetime predictions may be a viable and reliable alternative to specific decay date estimates.

Evaluation

While our models showed promise, their performance fell short of the high accuracy required for operational space debris management. The regression models' error margins of 6-7 years are substantial in the context of orbital decay prediction. The classification approach provided more reliable categorical predictions but still leaves room for improvement, especially for medium-lived objects.

Analysis

Several factors contributed to these outcomes:

1. **Data Limitations:** The lack of crucial variables like object mass and incomplete RCS data significantly hampered model performance.
2. **Domain Complexity:** The intricate physics of orbital mechanics and space debris behavior proved challenging to capture fully in our models without extensive domain expertise.
3. **Class Imbalance:** In the classification approach, medium-lived objects were consistently the most difficult to predict accurately, likely due to class imbalance and the inherent complexity of objects in this category.
4. **Model Strengths and Weaknesses:** Ensemble methods (Random Forest) and instance-based learning (KNN) outperformed other approaches, suggesting that the complex, nonlinear relationships in space debris decay patterns are better captured by these model types.

In conclusion, while our project did not achieve the level of predictive accuracy initially hoped for, it provided valuable insights into the challenges of space debris modeling and highlighted areas for future improvement. The results underscore the need for more comprehensive data collection, enhanced domain knowledge integration, and potentially more advanced modeling techniques to address this complex problem effectively.

Discussions

Future expectations from the project

Building upon our project's foundation, there are several promising avenues for future research and development that could significantly enhance the space debris decay prediction system. Our experience has illuminated key areas where focused improvements could yield substantial advances in prediction accuracy and model robustness.

A critical enhancement would involve implementing the mathematical modeling approach outlined in our project documentation. This would entail creating a simulated environment with approximately 400 satellite scenarios and 27 known debris cases, incorporating crucial parameters such as mass, size, and detailed orbital parameters. The mathematical model would generate precise decay date calculations with all dependency attributes, providing a rich dataset for training. This synthetic data could then be leveraged through transfer learning strategies, particularly using advanced neural network architectures like RNNs, LSTMs, or Transformers, which excel at time-series predictions.

Data acquisition and quality represent another significant area for advancement. Future teams could establish partnerships with space organizations to access more comprehensive datasets, including crucial features like precise RCS measurements, mass data, and cross-sectional areas. This would address one of the primary limitations we encountered with publicly available data. Additionally, implementing a systematic approach to real-time data collection and processing would ensure model currency and reliability.

The integration of advanced computational resources, particularly cloud computing platforms, could significantly expand the model's capabilities. This infrastructure would enable the implementation of more sophisticated machine learning techniques and allow for more extensive model training and validation. The increased computational capacity would also facilitate the exploration of ensemble methods and more complex neural network architectures.

Domain expertise integration represents another crucial area for development. A deeper understanding of orbital mechanics and space physics could inform more sophisticated feature engineering and model architecture decisions. This knowledge would be particularly valuable in developing and validating the mathematical model component of the system.

These future directions build upon our initial work while addressing key limitations we identified. By combining improved data quality, advanced modeling techniques, and robust computational infrastructure, future iterations of this project could achieve significantly higher prediction accuracy and reliability.

Challenges

There were many difficulties with this project. Obtaining the necessary data was one of the first significant challenges we faced. Public databases on artificial satellites and space debris are scarce, and the process of extracting data from Space-Track.org, the primary source we located, proved to be taxing. After collecting data all night, we discovered that several of the essential attributes we required were missing. We eventually moved to their API, which was significantly faster, but the problem of missing data persisted. Ultimately, it was difficult to train any significant model because over half of the records lacked the target variable.

Unfortunately, we could not access other data sources for satellite and debris information. Also, the research papers we found were incredibly detailed and technical — understanding them required a level of domain knowledge that we didn't have. We realized that applying complex mathematical models for orbital mechanics is not our cup of tea. Space debris is influenced by a range of forces, and while we understood that in theory, translating that into a machine learning model happens to be a different story.

Determining which criteria were truly important for forecasting the duration of debris' orbit was another challenge. Decay rates are known to be impacted by factors like solar drag, but it was challenging to determine how much and account for it in the model. We found it difficult to create a logical mathematical model without a firm grasp of the physics underlying it all. Furthermore, most of the data we extracted lacked crucial fields, such as radar cross-section (RCS) values or decay dates, which reduced the accuracy of our forecasts.

It was difficult to train the model on insufficient data. The results weren't as accurate as we had wanted, even after thorough preprocessing. We spent a lot of time fine-tuning the model to get its performance up to a point where the RMSE was barely acceptable, but in the end, the missing data held us back.

Overall, while the experiment didn't go as well as we'd expected, we learnt a lot about the complexity of space debris prediction. The difficulties we encountered highlight the need for improved data, additional knowledge, and a more sophisticated method of simulating these orbital patterns.

Conclusion

While this project evolved from its initial goal of developing a highly accurate AI/ML system for space debris prediction, it has delivered valuable insights and foundations for future work in this field. Our implementation of automated data collection and both regression and classification approaches (achieving 79.34% accuracy for classification, RMSE of 2335.84 days for regression with Random Forest models and RMSE of 2493.90 days for the tuned neural network) demonstrated both the potential and current limitations of AI in space debris management. Key challenges identified include insufficient data quality, particularly regarding mass and RCS measurements, and the complexity of orbital mechanics requiring sophisticated mathematical modeling. Moving forward, recommendations include developing synthetic data generation capabilities, implementing transfer learning approaches, and integrating deeper domain expertise. Although we didn't achieve the desired predictive accuracy, this project has provided ERETS and future researchers with crucial insights for resource allocation and realistic timeline establishment in space debris monitoring projects, contributing to the broader goal of sustainable space exploration.

References

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Appendix

Code and Project Management

We used Jira as the Project management tool and GitHub for version controlling.

Note:

- We are using a private repository. Please request for access if required (<https://github.com/ragwap/ERETS.git>)
- We were working together on a shared Colab notebook, so we didn't have to version control specifically. But we added our final code to the repo just to meet the guidelines.

The image displays two screenshots related to project management and code version control.

The top screenshot shows the Jira interface for a project named "ERETS Capstone". The project is in the "List" view, showing a table of tasks with columns for Type, Key, Summary, Status, Category, Assignee, Due date, and Priority. The tasks are listed as follows:

Type	Key	Summary	Status	Category	Assignee	Due date	Priority
Task	EC-1	Go through the repositories provided by ERETS - 1	DONE		Ashwin S		High
Task	EC-2	Go through the repositories provided by ERETS - 2	DONE		Truc Nguyen		High
Task	EC-3	Investigate the Research Papers Provided by ERETS - 1	DONE		Rmit370		High
Task	EC-4	Investigate the Research Papers Provided by ERETS - 2	DONE		Ravindu Anjana		High
Task	EC-5	Develop a web scraping tool for data collection	DONE		Truc Nguyen		Medium
Task	EC-6	Pre-process the data and flatten all the OMM data to a single	DONE		Ravindu Anjana		Medium
Task	EC-7	Principal component analysis (PCA)	DONE		Ashwin S		Medium
Task	EC-8	Develop a simple MLP to compare with regression models	DONE		Rmit370		Medium
Task	EC-9	Develop 3 regression models and train the dataset	DONE		Ravindu Anjana		Medium
Task	EC-10	Develop 3 regression models and train the dataset	DONE		Truc Nguyen		Medium
Task	EC-11	Add more layers and improve the neural network by fine-tuning	DONE		Ashwin S		Medium

The bottom screenshot shows the GitHub interface for the "ERETS" repository. The repository is private and contains 9 commits. The files listed are:

- .ipynb_checkpoints
- .gitignore
- ERETS.ipynb
- README.md
- code.ipynb
- data_with_RCS_value.jsonl
- enriched_satellite_data.jsonl
- get_rcs.py
- get_space_track_data.py
- merged_result.jsonl
- space_track.jsonl.zip

The repository also includes a README file and a list of contributors: xtrutt and ragwap.

Roles and Responsibilities

Name	Responsibilities
Vivek	• Contributor to the development of neural networks and its fine tuning • Contributor to pipeline of the project methodology • Contributor to the proposal and final reports
Truc	• Contributor to data collection tasks • Contributor to part of the data preprocessing • Contributor to the proposal and final reports
Ravindu (S3995505)	• Contributor to data loading and pre-processing. • Contributor to training and fine-tuning regression models. • Contributor to the proposal and final reports.
Ashwin	• Contributed to deciphering domain specific features and feature selection. • Contributed to training and fine-tuning neural networks. • Contributed to proposal and final report.

Self-Reflection | Vivek

This project has been more than just a part of the course for me. I have been fascinated by space all my life and learning that I could contribute towards sanitizing it from unwanted debris is a low-key achievement. In the beginning of the project, Truc, the data engineer in our team of 4, was able to extract data from space-track.org single handedly. I learned how to fetch and populate data from APIs of a website, in theory if not practically, nonetheless. Data extraction as step one was followed by gaining domain knowledge through the research papers provided to us via ERETS. The research papers I found to be quite extensive, rich in knowledge. I learned what the technical terms stand for, how the parameters influence the decay rate and the trajectory of a heavenly body, etc.

Our project's aim evolved with time, from making a mathematical model to predict and cross-verify the decay date and point of re-entry, to predicting the decay date using machine learning, which was more in our field of expertise. Lengthy discussions with our partner helped us clear up our doubts, and they successfully guided us to work towards a more achievable target. After the data-preprocessing, we started with predictive modelling. All the steps were discussed in a team session amongst our team members, and Ashwin, being our team leader, successfully managed and divided the roles to keep the project proceeding

according to the timeline. Coming to the modelling, the results of our models were not as expected. We switched to neural networks, as our next option. Unfortunately, the best error margins we got were 1300 days (about 3 and a half years) on average (which is quite large), better from predictive modelling, yet still far from optimal results. Looking to improve, we tried rectifying the data we had and searched alternate data sources. The original dataset had most of the crucial fields like RCS and date of decay missing, which we thought might be influencing our results by a substantial factor. In retrospect, higher computing power might have been useful as we could re-run the models repeatedly after every minor tweak, which was not possible in our case. Perhaps cloud computing would have been an excellent solution to this problem, however, it comes with a cost which might be unwise to pay without the guarantee of suitable results.

Summing up, this project led me to understand why cleaning up the space around Earth is so vital. Teamwork, time management, professionalism, were also contributors to my learning outside of academics. I wish we had more time to understand the complete working and practical use of our project. It has genuinely been exciting.

Self-Reflection | Truc

Through my involvement in this project, I gained valuable experience and learned several important lessons, particularly in my role leading the data collection efforts. Being responsible for this foundational component made me acutely aware that subsequent project phases depended on the quality and timeliness of my work - a situation commonly found in professional environments where different teams rely on each other's deliverables.

Initially, I approached web scraping using traditional browser automation methods, simulating user interactions to extract data from page sources. However, after encountering some challenges, I discovered that the target site offered an API endpoint for end-users. This discovery proved transformative – what previously took several hours could now be accomplished in just 3-5 seconds. This experience taught me the importance of thoroughly investigating available resources and APIs before implementing more complex solutions. In a professional setting, this lesson will help me optimize development time and resource utilization by first exploring existing solutions and tools before building custom implementations.

Another crucial lesson emerged regarding project scoping. We faced challenges due to insufficient initial clarity about the mathematical model requirements and their role in our prediction model development. This led us to spend considerable time trying to understand complex mathematical concepts without having the necessary domain knowledge to fully grasp all parameters and formulas. With the understanding and support of Dr. Ke and our project partner, we were able to successfully redefine the project scope and deliverables to better align with our capabilities and timeline. This experience has prepared me for the corporate environment where clear scope definition and effective stakeholder communication are essential. These lessons will be particularly valuable in my future career as they align with key professional competencies: efficient resource utilization, stakeholder management, and adaptive problem-solving. The experience has also highlighted the importance of balancing technical implementation with practical business constraints, a crucial skill in any technical role.

Self-Reflection | Ashwin

As I reflect on leading the space debris decay prediction project, I'm struck by how it embodied both the challenges and learning opportunities inherent in real-world data science applications. The project's complexity pushed me out of my comfort zone, particularly in dealing with limited and imperfect data. Working with categorical RCS values instead of precise measurements taught me valuable lessons about adapting modeling strategies to data constraints – something that theoretical coursework rarely prepares you for.

The experience of leading the team was particularly enlightening. While I had technical project management experience before, this project's unique challenges in communicating complex technical concepts and coordinating team efforts helped me grow as a leader. I learned to balance being decisive with remaining open to team input, especially when navigating through technical uncertainties. The need to maintain team morale while facing data quality issues and model performance challenges was particularly instructive.

Perhaps the most significant learning came from our inability to implement the partner's suggested mathematical modeling approach for transfer learning. This limitation, stemming from time and knowledge constraints, taught me the importance of early and honest assessment of team capabilities against project requirements. While our model's performance wasn't optimal due to data quality issues, this experience highlighted the critical relationship between data quality, domain knowledge, and model success. It also emphasized the importance of setting realistic expectations and being transparent about project limitations.

The project reinforced my understanding of feature engineering's critical role in machine learning projects. Working with incomplete data pushed us to be creative with feature selection and transformation, though ultimately it also showed us the limits of what we could achieve without certain crucial variables. This experience has made me a stronger advocate for comprehensive data collection and quality assurance in future projects, as I now deeply understand how upstream data decisions impact downstream modeling success.

Looking back, while we did not achieve all our initial objectives, the project provided invaluable lessons in real-world data science application, team leadership, and project management. These insights will undoubtedly influence my approach to future projects, particularly in terms of initial project scoping and data quality assessment.

Self-Reflection | Ravindu

As I reflect on my involvement in this project, I'm struck by the depth of learning and growth I experienced. My contributions to the development of regression models and the data loading and preprocessing stages were particularly enlightening. These tasks challenged me to think critically about how we handle and prepare data for complex AI applications, especially in a field as intricate as space debris prediction.

Working on the data loading and preprocessing phase was eye-opening. I gained a deep appreciation for the importance of data quality and structure in machine learning projects. Dealing with nested JSON structures and flattening them into a usable format taught me valuable skills in data manipulation. The

process of cleaning the data, handling missing values, and creating new features like 'LIFETIME_DAYS' reinforced the critical nature of thorough data preparation in AI projects.

Developing regression models was both challenging and rewarding. It was fascinating to see how different algorithms performed on our dataset, from simple linear models to more complex ensemble methods like Random Forest. The process of model selection and evaluation deepened my understanding of machine learning techniques and their applicability to real-world problems.

Writing the "Results, Evaluation & Analysis" section of the report was particularly insightful. It forced me to think critically about our models' performance and the broader implications of our work. Analyzing the strengths and weaknesses of our approaches, especially the challenges we faced with prediction accuracy, gave me a more nuanced understanding of the complexities involved in AI applications for space technology.

This project also highlighted the importance of domain knowledge in AI development. Our struggles with incorporating complex orbital mechanics into our models showed me that effective AI solutions often require a deep understanding of the field they're applied to.

Overall, this experience has been invaluable in bridging the gap between theoretical knowledge and practical application in AI. It has prepared me for future challenges in my career, teaching me the importance of adaptability, critical thinking, and continuous learning in the rapidly evolving field of artificial intelligence.

Figures

Figure 1: PCA Scree Plot

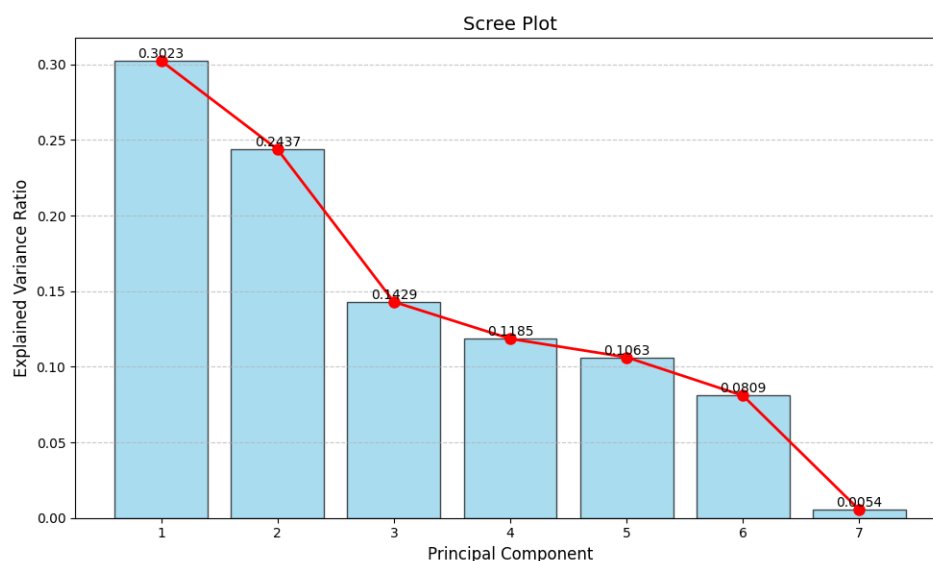


Figure 2: Neural Network Evaluation

